Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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A fixed-width language model has only d independent coordinates, yet at fixed angular tolerance it can geometrically accommodate exponentially many nearly orthogonal feature directions representing "meaning" and semantics. Lévy concentration and Johnson–Lindenstrauss embeddings anchor this claim, but coexistence does not guarantee cost-free simultaneous readout. For uniformly random directions, a typical pair overlaps by order $1/\sqrt{d}$; if both features are active with similar strengths, an inner-product readout of either receives an unwanted contribution of erroneous meaning of about $1/\sqrt{d}$ times the desired signal. Separately, we also discuss encoding polysemy by quantum inspired intertwining contexts; that is, contexts sharing one or more elements.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume's broader inquiry into "quantum thinking." A short operational picture is enough to begin. Text is divided into tokens; each token is mapped to an initial vector; Transformer layers turn those initially context-free vectors into context-dependent hidden states; and an output map assigns probabilities to possible next tokens [1]. Section II explains this loop in plain language. Appendix A gives an explicit formal version, with every quantity defined, that may be skipped on a first reading.

The central question is simple: how can a space with only d independent coordinates accommodate a much larger dictionary of approximately distinct semantic directions, and what prevents all of them from being used at once? The positive thesis has two parts. High dimension makes a large dictionary geometrically possible, but useful meaning is limited by coactivation and readout. Learning should therefore spend its limited angular separation where cross-talk is costly—between important features that often activate together. For lexical ambiguity, a reusable token component may additionally be shared across several low-rank sense models.

This is an analytical and model-proposal chapter; it reports no new experiment. The geometric results are established mathematics used as null benchmarks. The coactivation prediction and the tied shared-subspace model are empirical hypotheses, and Sec. VII D gives tests under which they fail. Stating that status early is important: a possible arrangement of vectors is not evidence that an LLM has learned that arrangement.

a. Elementary mathematical map. Most steps below reduce to familiar undergraduate facts. The Pythagorean theorem and the orthogonal projection theorem split a vector into perpendicular components. The Cauchy–Schwarz and triangle inequalities bound inner products and accumulated error. The polarization identity converts squared distances into inner products. The spectral theorem diagonalizes real symmetric matrices. Finally, additivity of variance explains root-mean-square growth, and the union bound says that the chance of any bad event is at most the sum of their individual chances. The core argument uses these elementary facts. Lévy concentration and the Johnson–Lindenstrauss (JL) lemma are quoted; a few explicitly optional comparisons use additional standard results and may be skipped.

Let d be the number of vectors in an orthonormal basis of the usable carrier. It is the one exact vector-space dimension in the chapter. Let M be the measured number of feature directions returned by a declared extraction and duplicate-removal rule. It is a count, not a dimension. For unit vectors, Cauchy–Schwarz gives $|u^\mathsf{T} v| \le 1$, and $u^\mathsf{T} v$ is the cosine of their angle. We call a dictionary ε-quasi-orthogonal when every absolute pairwise inner product is at most the chosen tolerance $0 < \varepsilon < 1$. The function $M_\varepsilon(d)$ is the largest possible size of such a finite code in $\mathbb{R}^d$; it is not a second dimension.

High-dimensional concentration and JL projection both lead to the same single-exponential existence scale: for fixed ε and sufficiently large d, one can construct at least $\exp(c\varepsilon^2 d)$ quasi-orthogonal directions for some constant $c > 0$. These are related formulations of one concentration phenomenon, not two independent capacity mechanisms. JL preserves a prescribed finite geometry; random spherical coding instead supplies a null distribution for a sampled dictionary and its most extreme overlap. A conditional double exponential after separately assuming a fully accessible tensor-product carrier is discussed separately in the conditional remark in Sec. III C; it is not an LLM capacity claim.

Geometric coexistence is only the first half of the problem. Suppose k random-like features are active together. A simple correlation readout of one chosen feature receives many small cross terms. Their variances add, giving a root-mean-square (RMS) interference scale $\sqrt{k/d}$. This is the Pythagorean rule applied to random errors: it describes average squared fluctuation, not cancellation

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in every example. The empirical prediction is therefore about the placement of overlap, not merely its existence. Important, frequently coactive features should cause less held-out cross-talk than activity-matched controls.

Lexical ambiguity supplies the second hypothesis. The same token can occur in financial, river-edge, or seat/bench uses of "bank," while its surrounding sequence drives different contextual states. The proposal ties a token-associated reference subspace across senses and learns separate low-dimensional residual subspaces. Tying pools evidence for the common component and prevents each sense model from relearning it. The predicted benefit is regularization and sample efficiency, especially for rare or imbalanced senses; with abundant data or no stable shared component, the constraint may do nothing or may hurt. Because every vector admits an orthogonal decomposition, merely finding a shared component is trivial. The nontrivial test is incremental held-out value beyond token-mean subtraction, top-principal-component removal, and parameter-matched unstructured models.

Current Transformers remain classical systems. Real vectors, inner products, high dimension, or probabilistic output do not make a computation quantum. Here "quantum-inspired" refers only to a possible organization by context-indexed projectors and symmetric operators. The operator vocabulary earns a specifically quantum-like role only if prespecified sequential probes predict order effects beyond equally expressive classical baselines. The relation to quantum cognition and compositional distributional semantics is reviewed in Sec. VII B; no physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

A practical caveat concerns anisotropy: an LLM layer may use fewer directions than its nominal width suggests. Here d means the *usable exact dimension* after a declared restriction or whitening rule. By the spectral theorem, the empirical covariance matrix has an orthonormal eigenbasis; keeping or rescaling selected eigenvectors defines the tested carrier, and d counts those retained axes. The resulting numerical rank is protocol-dependent, so the selection rule must be reported.

The lexical hypothesis is intended for ambiguous tokens with established inventories such as WordNet or OntoNotes and enough labeled contexts from resources such as SemCor or the Word-in-Context (WiC) benchmark [2]. It is adjacent to the superposition hypothesis in mechanistic interpretability, according to which neural networks may represent more features than neurons by using nonorthogonal directions with sparse activity [3]. The present chapter separates that global geometric background from a local, testable organization of polysemy.

The remaining sections follow the argument in that order. Section III develops the geometric null model; Sec. IV turns coexistence into a readout problem; Secs. V and VI present the tied shared-subspace model and its router; and Sec. VII relates the proposal to existing work, limitations, and proposed experiments.

For reference, $\mathbb{R}^d$ is the usable Euclidean carrier, $u^\mathsf{T} v$ denotes the inner product, the superscript T denotes transpose, and $\|v\| = \sqrt{v^\mathsf{T} v}$. The unit sphere is $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$; I is the identity matrix; $\mathbb{P}(A)$ is the probability of an event A, $\mathbb{E}[Y]$ is the average of a random quantity Y, and $\mathrm{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2]$ is its mean squared fluctuation. The symbols $\approx$, $\sim$, $\lesssim$, and $\gtrsim$ below indicate scaling or approximation at the stated level, not exact equality. Local ranks and error tolerances are introduced only where they are used.

II. FROM TEXT TO WEIGHTS AND BACK TO TEXT: THREE COMPUTATIONAL LOOPS

The word *weight* is used for two different objects in informal accounts of Transformers. A trained model has persistent numerical parameters: its embedding table, query, key, value, output, and feed-forward matrices, normalization parameters, and final vocabulary readout. Self-attention also computes input-dependent coefficients that are often called attention weights. Those coefficients exist only for the current forward pass. They do not become persistent model parameters. Keeping this distinction visible separates the three computational loops below.

A. Creating the weights: tokenization, contextual vectors, and training

Training begins with text, but a neural network does not receive words or sentences as primitive objects. A tokenizer first segments character strings into a finite vocabulary of token types and replaces each occurrence by an integer identifier. Tokens may be whole words, word fragments, punctuation, spaces, or byte-level units. Modern subword schemes are designed to cover rare and previously unseen words without assigning a separate vocabulary entry to every possible word [4, 5]. The tokenizer is normally learned or chosen before the main language-model optimization and is then held fixed during that optimization.

An embedding table associates each token identifier with an initial real vector. At this first lookup stage, two occurrences of the same token use the same table entry. Position information is also supplied, either through added position vectors or through mechanisms such as rotary transformations inside attention; some systems add role or segment information as well. The lookup vector is therefore not yet the fully contextual meaning of that occurrence. Context dependence is created by the successive Transformer layers.

Within one self-attention head, the current vector at each position is mapped by learned matrices to a query, a key, and a value. The query at one position is compared by an inner product with keys at other permitted positions. In the decoder-only autoregressive Transformer considered here, a causal mask excludes future positions during next-token training. Softmax normalization turns the permitted comparisons into nonnegative attention coefficients, and those coefficients sum to one. The update is therefore a weighted average of the value vectors. Cauchy–Schwarz bounds each raw query–key comparison by the product of the two vector lengths; the learned matrices determine which directions make that comparison large. Multiple heads can learn different patterns of dependence. Residual connections, normalization, and feed-forward blocks then transform the update further. After many layers, the hidden vector for an occurrence of "bank" can differ substantially between financial, river-edge, and seat/bench contexts even though all occurrences began from the same token-table entry [1].

This forward calculation still does not create new persistent weights. The last hidden state at each training position is converted into scores for the next token. The discrepancy between the predicted distribution and the observed next token defines a training loss. Backpropagation differentiates that loss through the output map, feed-forward blocks, attention operations, and embedding table. Mathematically, backpropagation is repeated application of the multivariable chain rule. An optimizer then changes the persistent parameters. Repeating this cycle over many batches creates the trained weight values. In short, self-attention produces contextual activations and differentiable routes through which error signals flow; gradient-based optimization produces the persistent model weights [6].

Does this description require a Hilbert space? Nothing here requires more than finite-dimensional Euclidean inner-product geometry. Every finite-dimensional Euclidean space is complete and may therefore formally be called a real Hilbert space, but completeness plays no computational role and implies no quantum mechanism. The construction needs ordinary real vectors, inner products, matrices, and nonlinear maps; it does not require complex amplitudes, Born probabilities, or quantum hardware. Nor does training assign one exactly orthogonal vector to every token or every meaning. The quasi-orthogonal geometry studied later concerns a declared dictionary of extracted feature directions at a chosen resolution. It is a possible organization of learned semantic distinctions, not an assumption built into tokenization or self-attention.

B. Generating an output: tokenization and autoregressive feedback

Response generation begins with the same tokenizer. The prompt, including any control, role, or formatting tokens exposed to the model, is converted to a token sequence and then to initial vectors. The trained parameters are now normally fixed. A causal forward pass produces contextual hidden states for the prompt, and the final relevant state is mapped to one score for every token in the vocabulary. Softmax converts these scores into a next-token distribution. Greedy decoding chooses its largest entry; sampling may use a temperature or restrict attention to a high-probability subset before drawing a token.

The selected token is appended to the sequence. The enlarged sequence is then the context for the following prediction, so the generated token can change all subsequent attention patterns and hidden states. This loop is repeated until an end condition is reached. Implementations commonly cache previous keys and values to avoid recomputing unchanged parts of the prefix, but caching changes efficiency rather than the trained parameters. Finally, the tokenizer's inverse assembly rules turn the generated token identifiers back into a character string [5].

Autoregressive generation is thus already recursive in a precise but limited sense: each output token becomes part of the next input. The context, attention coefficients, and hidden vectors evolve from step to step, while the persistent model weights remain fixed. The standard decoding interface does not require a separately represented complete answer; it emits tokens successively by repeatedly mapping the current vector-valued context to a distribution over the next discrete token.

C. Recursive adaptation, alignment, and hallucination

Three forms of recursion should be separated. First, in autoregressive generation each output token becomes part of the next input, while the model parameters remain fixed. Second, an external memory, retrieval result, tool output, or longer conversation history can enlarge the next input without changing those parameters. Third, genuine online learning can update selected parameters after an interaction. Dynamic evaluation shows that the third loop is technically possible [7], but it is a separate learning rule rather than ordinary frozen-weight inference.

This subsection is orientation only; no later geometric conclusion depends on it. Online learning needs verified feedback and safeguards against drift, forgetting, poisoning, and error reinforcement. Post-training alignment is also a separate intervention: it changes the objective toward selected forms of instruction following, helpfulness, and safety [8]. Alignment and factual accuracy are not opposite ends of one scale. Preference optimization can encourage agreement in some settings [9], while hallucination also occurs without alignment [10].

Whether a particular alignment procedure increases or decreases unsupported claims is therefore a causal question, not a consequence of the word "alignment." It requires a matched control with the same base model, data exposure, compute, prompt distribution, grounding, and decoding rule. Appendix A 3 gives the optional update equation, the relevant safeguards, and this matched comparison in more detail.

III. HOW HIGH DIMENSION SUPPORTS MANY NEARLY ORTHOGONAL DIRECTIONS

The question is now geometric: at a fixed angular resolution, how does the number of available directions grow with the exact carrier dimension? Here d is the usable carrier rank, whereas M counts the directions in one measured dictionary and may exceed d. For a dictionary

$$W = \{w_1, \dots, w_M\} \subset S^{d-1}, \qquad \mu(W) = \max_{i \neq j} |w_i^\mathsf{T} w_j|. \tag{1}$$

For $0 < \varepsilon < 1$, we call this particular dictionary ε-quasi-orthogonal when $\mu(W) \leq \varepsilon$, and write $M_\varepsilon(d)$ for the maximum cardinality achievable in $\mathbb{R}^d$ under this condition. Thus $M_\varepsilon(d)$ and the empirical count M are cardinalities, not additional vector-space dimensions. The term *quasi-dimensionality* will refer only to this resolution-dependent abundance of directions.

Two elementary facts place this definition. Exactly orthogonal nonzero vectors are linearly independent, so rank–nullity limits an exactly orthogonal family to d members. Quasi-orthogonal vectors need not be independent. Nevertheless, $M_\varepsilon(d)$ is finite when $\varepsilon < 1$: the condition keeps every pair a fixed positive distance apart, while the unit sphere (with opposite signs identified) is compact. A compact set cannot contain infinitely many points separated by the same positive distance. Thus this is an ordinary finite packing problem, not a new linear dimension.

The mathematical ingredients below are established results. Lévy concentration says that near-orthogonality is typical relative to a fixed direction. The JL lemma repackages the same high-dimensional concentration as a finite-set projection theorem. Direct random coding supplies a null ensemble for the largest overlap in a sampled dictionary. These are related views of one phenomenon, useful for different diagnostics; their rates are not independent effects to be multiplied. None identifies semantic features in an LLM.

A. Lévy concentration: why random directions are nearly perpendicular

In one standard form of Lévy's lemma, let $f : S^{d-1} \to \mathbb{R}$ be L-Lipschitz: changing its input by a distance r changes its value by at most Lr. If m_f is a median of f, then, for $t > 0$,

$$\mathbb{P}\{|f - m_f| \geq t\} \leq 2 \exp\left[-c\,\frac{(d-1)t^2}{L^2}\right] \tag{2}$$

for a universal constant $c > 0$ [11, 12]. The choice of median or mean and the numerical constant vary among formulations; the invariant content is exponential concentration in dimension for every Lipschitz scalar probe at fixed t/L.

Fix $u \in S^{d-1}$ and take $f_u(x) = u^\mathsf{T} x$. This function is 1-Lipschitz because Cauchy–Schwarz gives $|u^\mathsf{T}(x-y)| \leq \|x-y\|$. By rotational symmetry it has median and mean zero. For uniform $x \in S^{d-1}$,

$$\mathbb{E}[u^\mathsf{T} x] = 0, \qquad \mathrm{Var}(u^\mathsf{T} x) = \frac{1}{d}. \tag{3}$$

The variance $1/d$ is just Pythagoras plus symmetry. Rotate coordinates so that u is the first basis vector. Since $x_1^2 + \cdots + x_d^2 = 1$ on the unit sphere and the coordinates have identical distributions, each has expected square $1/d$. Chebyshev's inequality would already show that large overlaps become uncommon, but only with a polynomial bound. Lévy's lemma sharpens this to the exponential spherical-cap estimate

$$\mathbb{P}\{|u^\mathsf{T} x| \geq \varepsilon\} \leq 2 \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right], \qquad 0 < \varepsilon < 1. \tag{4}$$

Thus almost every direction lies close to the equator perpendicular to a specified direction, with a typical overlap amplitude of order $d^{-1/2}$.

There is an equivalent angular description. If the sign of a direction is irrelevant, its principal angle from u is $\theta_{\mathrm{pr}} = \arccos|u^\mathsf{T} x|$. The first-order Taylor approximation near zero is $\arccos z \approx \pi/2 - z$, with only cubic and higher corrections, so its typical deviation from $\pi/2$ is of order $d^{-1/2}$. In the independently sampled isotropic null, a randomly chosen pair therefore exhibits *angular homogenization*: its principal angle lies in a band of width proportional to $d^{-1/2}$ near $\pi/2$ at any fixed confidence level. Increasing the dictionary size at fixed d does not narrow this typical-pair distribution; instead it increases the extreme overlap, as quantified below. High dimension creates both the typical angular concentration and room for exponentially large finite codes.

a. Optional complex comparison. For comparison with the later operator language, the corresponding complex statement is especially sharp. If ϕ, ψ are independent unit vectors drawn uniformly under all unitary rotations (the meaning of "Haar-random"), then $Z = |\langle \phi, \psi \rangle|^2$ has the $\mathrm{Beta}(1, d-1)$ distribution. Using the same ε for an overlap *amplitude*, rather than changing midway to a squared-overlap tolerance, gives the exact tail

$$\mathbb{P}\{|\langle \phi, \psi \rangle| \geq \varepsilon\} = (1 - \varepsilon^2)^{d-1} \leq \exp[-(d-1)\varepsilon^2]. \tag{5}$$

Unitary symmetry lets us hold ϕ fixed as the first coordinate vector. The squared coordinate magnitudes sum to one, the complex analogue of Pythagoras, and integrating the first-coordinate density gives the displayed tail. Readers who wish to remain in real vector spaces may skip this exact comparison; no later LLM claim depends on it.

The real and complex formulas differ in constants and exact distributions, but both exhibit the same operative exponential scale $\exp[-cd\varepsilon^2]$.

B. Johnson–Lindenstrauss: preserving a finite cloud of points

Lévy's lemma describes most points on a sphere. The JL lemma gives a related finite-set formulation: a finite cloud can be compressed while all pairwise squared distances change by at most a chosen relative error. The surprising feature is that the required target dimension grows only as the logarithm of the number of points. Specifically, for $0 < \delta < 1$ and a finite set X, there is a linear map Φ into $\mathbb{R}^d$ such that

$$\begin{aligned}(1-\delta)\|x-y\|^2 &\le \|\Phi x - \Phi y\|^2 \\ &\le (1+\delta)\|x-y\|^2, \qquad x, y \in X,\end{aligned} \tag{6}$$

provided

$$d \ge C_{\mathrm{JL}} \delta^{-2} \log |X| \tag{7}$$

for a universal C_{JL} [13, 14].

To make the connection to quasi-orthogonality explicit, let e_i be the standard coordinate vector with a one in position i and zeros elsewhere, and take

$$X = \{0, e_1, \ldots, e_M\} \subset \mathbb{R}^M.$$

Including the origin is important: Eq. (6) then controls both the norms $\|\Phi e_i\|$ and the distances $\|\Phi e_i - \Phi e_j\|$. The real polarization identity gives

$$2(\Phi e_i)^\mathsf{T}(\Phi e_j) = \|\Phi e_i\|^2 + \|\Phi e_j\|^2 - \|\Phi e_i - \Phi e_j\|^2. \tag{8}$$

This identity follows by expanding the three squared norms and cancelling terms. In the original set, $\|e_i\|^2 = 1$ and $\|e_i - e_j\|^2 = 2$. Because the origin was included, JL controls each projected norm as well as each projected distance. Substituting those bounds into the polarization identity gives $|(\Phi e_i)^\mathsf{T}(\Phi e_j)| \le 2\delta$, while $\|\Phi e_i\|^2 \ge 1 - \delta$. After normalization, $w_i = \Phi e_i / \|\Phi e_i\|$, one obtains

$$|w_i^\mathsf{T} w_j| \le \frac{2\delta}{1-\delta} \qquad (i \ne j). \tag{9}$$

Normalization divides by $\|\Phi e_i\|\|\Phi e_j\| \ge 1 - \delta$, which explains the denominator. Taking $\delta = \varepsilon/(2+\varepsilon)$ makes the right-hand side of Eq. (9) equal to ε. Reading Eq. (7) constructively therefore gives, for every fixed $0 < \varepsilon < 1$ and all sufficiently large d,

$$M_\varepsilon(d) \ge \exp(c'\varepsilon^2 d), \tag{10}$$

for a universal $c' > 0$. This is an existence lower bound. Sharp lower bounds for worst-case JL dimension reduction [15, 16] do not, by themselves, establish a matching universal upper bound for spherical codes, and no such upper bound is claimed here. Algebraically, the exponential appears simply by solving the logarithmic dimension condition for the number of points: undoing a logarithm requires exponentiation.

C. Direct random coding and the Welch floor

The exponential scale can also be reached directly. Draw the M directions independently from the uniform measure on S^{d-1}. Applying Eq. (4) to all unordered pairs gives

$$\mathbb{P}\{\mu(W) \ge \varepsilon\} \le M(M-1) \exp\left[-\frac{(d-1)\varepsilon^2}{2}\right]. \tag{11}$$

There are $\binom{M}{2}$ unordered pairs. The union bound says that the probability that at least one pair is bad is no larger than the sum of the individual bad-pair probabilities; the pair events need not be independent. The factor two in Eq. (4) cancels the factor one-half in the number of pairs. Whenever the right-hand side is less than one, at least one ε-quasi-orthogonal dictionary must exist: failure cannot occur for every random draw. This is the elementary probabilistic method. It reproduces the $\exp(c\varepsilon^2 d)$ lower scale through the same underlying concentration mechanism as JL, now as an ensemble calculation rather than a finite-metric projection statement. The two rates must not be multiplied or counted as independent amplifications. Equation (11) is only a sufficient random-code bound; it is neither an optimized-code upper bound nor a semantic capacity theorem.

Equating the number of pairs to the real spherical-cap tail gives the useful finite-sample null estimate

$$\mu_{\max}^{\mathrm{null}}(d, M) \approx 2\sqrt{\frac{\log M}{d}}, \qquad \log M \ll d, \tag{12}$$

up to lower-order spherical-cap factors. The estimate comes from setting the right-hand side of the union bound to a number of order one. Taking logarithms turns a product into a sum, and solving $M^2 \exp(-d\varepsilon^2/2) \approx 1$ for ε gives the displayed square root. It is therefore an extreme-value calibration for a random ensemble. It describes the most extreme pair in an independently sampled isotropic dictionary, not the overlap of a typical pair and not a hard limit on a learned one.

A genuine deterministic restriction is supplied by the Welch bound:

$$\mu(W)^2 \ge \frac{M-d}{d(M-1)} \qquad (M > d), \tag{13}$$

for every unit-vector dictionary [17]. The Welch floor, the random maximum in Eq. (12), and the existence lower bound in Eq. (10) answer different questions and should not be presented as the two edges of one universal interval.

The Welch bound itself follows from familiar linear algebra. Form the Gram matrix $G = (w_i^\mathsf{T} w_j)_{i,j}$, whose entries are all pairwise inner products. It is positive semidefinite, has rank at most d, and has trace M. By the spectral theorem it has at most d nonzero eigenvalues, and Cauchy–Schwarz says that the sum of their squares is at least M^2/d. The diagonal of G already contributes M

to that sum of squares. The average squared off-diagonal entry must therefore be at least the right-hand side of Eq. (13); the maximum cannot be smaller than the average.

a. Optional complex code. The exact complex-Haar law in Eq. (5) makes the corresponding random-code statement particularly transparent. If $M_\varepsilon^{\mathbb{C}}(d)$ denotes the maximum cardinality in $\mathbb{C}^d$ at the same amplitude tolerance, a union bound over all pairs gives the explicit existence result

$$M_\varepsilon^{\mathbb{C}}(d) \geq \left\lfloor \exp\left[\frac{d-1}{2}\big(-\log(1-\varepsilon^2)\big)\right]\right\rfloor. \tag{14}$$

This is the sharper complex analogue of the real exponential scale, not an additional linear dimension.

b. Conditional tensor-product remark (not an LLM capacity claim). The conditional double exponential requires two separate steps. If an independently justified carrier is the full tensor product $\bigotimes_{a=1}^n \mathbb{R}^q$, with fixed $q > 1$, then its exact dimension is

$$d = q^n. \tag{15}$$

The elementary tensor-product basis is indexed by n independent choices, each with q possibilities, so the multiplication principle gives q^n basis vectors. Quasi-orthogonal coding is then exponential in that already exponential dimension. For fixed $0 < \varepsilon < 1$, substitution into Eq. (10) gives

$$M_\varepsilon(q^n) \geq \exp\!\big(c'\varepsilon^2 q^n\big), \tag{16}$$

which is doubly exponential in n. Equation (14) gives the same conclusion, with an explicit exponent, for a full complex tensor-product carrier. These bounds concern arbitrary directions in the full carrier. They do not follow if admissible states are restricted to product vectors or another low-complexity manifold; access to the full carrier, or a code construction within the admissible class, is an additional premise. An ordinary Transformer residual stream is presented only as a measured real carrier $\mathbb{R}^d$. Its width, layers, heads, or token positions do not by themselves establish independent semantic tensor factors. For an LLM the unconditional geometric statement is therefore an existence lower bound exponential in the declared usable d; Eq. (16) is a further tensor-product hypothesis, not an architectural consequence. No later semantic conclusion in this chapter depends on this remark.

IV. FROM GEOMETRIC COEXISTENCE TO SEMANTIC READOUT

An exponential dictionary can coexist geometrically without all its coefficients being simultaneously accessible. Readout depends on which directions activate together, their amplitudes, the decoder, and the required error probability. This distinction is where the geometry acquires potential semantic content.

A. Fixed-target, worst-case, and joint readout

Let an active set S of size k produce the hidden state

$$h = \sum_{j\in S} x_j w_j. \tag{17}$$

Here S is the set of features active in this example, $k = |S|$, and x_j is the amplitude of feature j. This is simply a linear combination of the active directions. For one specified active target $i \in S$, a correlation decoder returns

$$w_i^\top h = x_i + \sum_{\substack{j\in S\\ j\neq i}} x_j w_i^\top w_j. \tag{18}$$

This identity uses only distributivity of the inner product and $w_i^\top w_i = 1$. The first term is the desired signal; every cross inner product in the remaining sum is interference. In the exact independently sampled isotropic ensemble, provided S and the amplitudes are fixed or selected independently of the direction draws, conditional on S, (x_j), and w_i, the variables $w_i^\top w_j$ for $j \neq i$ are independent, centered, and have variance $1/d$. Consequently

$$\operatorname{Var}\left[\sum_{\substack{j\in S\\ j\neq i}} x_j w_i^\top w_j \;\middle|\; S, (x_j), w_i\right] = \frac{1}{d}\sum_{\substack{j\in S\\ j\neq i}} x_j^2. \tag{19}$$

For independent centered random terms, covariance terms vanish, so the variance of a sum is the sum of the variances. This is the probabilistic version of Pythagoras. It describes average squared fluctuation and does not assert cancellation in each individual context. For a learned, merely random-like dictionary this equality is used only as an approximate null calibration. For comparable order-one amplitudes it gives the fixed-target RMS scale

$$\sigma_{\mathrm{int}} \sim \sqrt{\frac{k-1}{d}} \approx \sqrt{\frac{k}{d}}. \tag{20}$$

This estimate is a property of the declared correlation decoder under cancellation assumptions, not a universal recovery law.

Without cancellation, coherence yields the deterministic target-wise bound

$$|w_i^\top h - x_i| \leq \mu(W)\sum_{\substack{j\in S\\ j\neq i}} |x_j|. \tag{21}$$

This bound follows from the triangle inequality and $|w_i^\top w_j| \leq \mu(W)$. Unlike the RMS estimate, it assumes no cancellation, so the possible error grows linearly with the absolute amplitudes. The distinction becomes sharper for joint recovery. With W_S the matrix whose columns are the active directions, let x_S collect their amplitudes and let $\widehat{x}_S$ collect their correlation readouts. With $G_S = W_S^\mathsf{T} W_S$, simultaneous readout gives

$$\widehat{x}_S = W_S^\mathsf{T} h = G_S x_S, \qquad \|G_S - I\|_{\mathrm{op}} \le (k-1)\mu(W), \tag{22}$$

where $\|A\|_{\mathrm{op}}$ is the operator norm, the largest factor by which A can stretch a unit vector. The last inequality follows, conservatively, from Gershgorin's theorem. The Gram matrix has ones on its diagonal and pairwise overlaps off the diagonal. Gershgorin's theorem says that each eigenvalue lies within an off-diagonal row sum of a diagonal entry; here that row sum is at most $(k-1)\mu(W)$. If it is below one, G_S is invertible and coefficients on a known support can be recovered by applying G_S^{-1}. If $k > d$, rank–nullity instead makes G_S singular. Support identification must additionally control false readouts among the $M - k$ inactive directions, introducing a multiple-comparison cost. Uniform recovery of every sparse coefficient vector is therefore a stronger problem than the fixed-target calculation above: it requires additional assumptions and is not implied by the coherence or JL statements. We do not use such a uniform-recovery theorem here. The exponential coexistence of directions, fixed-target RMS access, worst-case access, and joint coefficient recovery are four distinct statements.

B. Why semantic interference must be conditioned on coactivation

Maximum coherence assigns the same cost to every pair. A semantic decoder does not: two directions that never activate together can overlap without contributing to the error in Eq. (18). Conversely, even a moderate overlap can be expensive when the pair coactivates often, carries large amplitudes, and matters to the task. Define the ordered, nonnegative activity-and-importance weight

$$a_{ij} := \omega_i \, \mathbb{E}\big[\mathbf{1}_{\{i,j \text{ active}\}} x_j^2\big], \qquad \omega_i \ge 0, \tag{23}$$

where ω_i is a nonnegative scalar recording the task importance of accurately reading feature i. The indicator $\mathbf{1}$ equals one when both features are active and zero otherwise. Thus a_{ij} is the importance of reading i, multiplied by the average squared amplitude of j on examples where the pair coactivates; it vanishes when they never coactivate. For a fixed target i, write the contribution from feature j as $c_{ij} = \mathbf{1}_{\{i,j \text{ active}\}} x_j (w_i^\mathsf{T} w_j)$. If distinct contributions have zero cross-moments on the held-out distribution, $\mathbb{E}[c_{ij} c_{ij'}] = 0$ for $j \ne j'$, then the diagonal part of the expected importance-weighted squared error is

$$E_{\mathrm{int}} = \sum_{i \ne j} a_{ij} (w_i^\mathsf{T} w_j)^2. \tag{24}$$

Squaring the interference sum produces diagonal terms with the same feature index and cross terms with two different indices. Equation (24) keeps the diagonal part; the cross terms vanish only under the stated zero-cross-moment assumption. We use E_{int} as a coactivation-weighted diagonal interference proxy. Without that assumption, the full squared error also contains cross-moment terms involving distinct j and j'. If the contributions are centered, these cross-moments are covariances.

This changes the scientific prediction. A useful learned dictionary need not beat the isotropic null on every pair, and a nonzero overlap is not itself a defect. If training is pressured to reduce semantic readout error, then important, frequently coactive pairs should contribute less to Eq. (24) than they do in controls matched for d, M, marginal activity, amplitude, and feature importance. That conditional redistribution, not overcompleteness alone, is testable.

This prediction extends rather than replaces earlier toy-model work on superposition, where correlated and anticorrelated features can favor different learned geometries [3]. The additional proposal here is the held-out, task-weighted statistic in Eq. (24) for extracted contextual features. The general observation that feature correlation affects packing is not claimed as new.

C. A task-dependent operating window

There is no task-independent numerical bracket whose lower edge is d and whose upper edge is a universal packing capacity. A meaningful operating window is instead the intersection of an empirical coverage requirement with declared readout tolerances. Let $\varepsilon_{\mathrm{task}}$ be the largest tolerated absolute pairwise overlap for a chosen diagnostic and let Δ be the tolerated fixed-target RMS interference. For the real isotropic null, Eqs. (12) and (20) give the separate crossover scales. They come from elementary algebra: set the random maximum $2\sqrt{\log M/d}$ equal to $\varepsilon_{\mathrm{task}}$, and set the RMS interference $\sqrt{k/d}$ equal to Δ. Solving for the two counts gives

$$M_{\mathrm{crowd}}(d, \varepsilon_{\mathrm{task}}) \approx \exp\!\left(\frac{d\varepsilon_{\mathrm{task}}^2}{4}\right), \qquad k_{\mathrm{rms}}(d, \Delta) \lesssim d\Delta^2. \tag{25}$$

The first approximation assumes the same small-overlap regime used in Eq. (12). The symbols $\approx$ and $\lesssim$ mark design scales, not sharp universal inequalities. The first concerns the worst of order M^2 pairs formed from M independently sampled directions; the second concerns one specified target among k active directions. Let $M_{\mathrm{req}}(\mathcal{T})$ be the smallest measured dictionary size that reaches a prespecified coverage or performance level on task $\mathcal{T}$, and let $E_{\max}(\mathcal{T})$ be its allowed value of the interference proxy. A useful task-conditioned summary is

$$\begin{aligned} M_{\mathrm{req}}(\mathcal{T}) &\le M, & \mu(W) &\le \varepsilon_{\mathrm{task}}, \\ k &\lesssim k_{\mathrm{rms}}(d, \Delta), & E_{\mathrm{int}} &\le E_{\max}(\mathcal{T}). \end{aligned} \tag{26}$$

Read from left to right, the display says: retain enough directions to cover the task; require the measured worst pairwise overlap to stay below its chosen tolerance; activate only as many directions as the RMS budget permits; and keep measured coactivation-weighted error below its task budget. Only the coverage threshold comes from semantic performance. The others are declared geometric or readout criteria. The scale M_{crowd} in Eq. (25) remains an isotropic-null diagnostic, not a substitute for measuring $\mu(W)$ and not a universal spherical-code upper bound. A learned dictionary can lie beyond that random scale and still satisfy the declared overlap tolerance. A stricter joint-recovery criterion may replace the fixed-target condition through Eq. (22). The Goldilocks region is thus the set of dictionaries that provide enough measured task coverage while satisfying explicitly selected pairwise, active-set, and readout-error criteria. Its lower edge comes from the task's coverage curve, not from an orthogonality count.

For scale, if $d = 4096$ and $M = 10^4$, Eq. (12) predicts a random-dictionary maximum of about 9.5%, although one typical pair has RMS overlap only $1/\sqrt{d} \approx$ 1.6%. Separately, $k = 64$ gives the fixed-target scale $\sqrt{k/d} = 0.125$. These numbers illustrate why the dictionary count, the most extreme pair, and simultaneous readout cannot be collapsed into one effective dimension.

Nothing in these benchmarks is specifically semantic until a feature extractor, activity distribution, decoder, and task are fixed. The next section adds a separate and stronger hypothesis about how context-dependent readout may be organized for lexically ambiguous tokens.

V. A SHARED-SUBSPACE MODEL OF LEXICAL AMBIGUITY

The previous section supplied classical geometric benchmarks. It did not imply any particular organization of lexical ambiguity. We now propose a separate classical model: occurrences of one token may share a reusable low-dimensional component, while context-specific variation occupies residual subspaces. The shared part can represent lexical identity, morphology, or a token-specific baseline. Tying it across senses pools training examples and reduces the number of separately estimated directions. The expected advantage is therefore regularization and sample efficiency, not greater expressivity. It should be largest for rare or imbalanced senses and can disappear or reverse when no stable shared component exists.

This supplies a possible bridge from global quasi-dimensionality to local meaning. A linguistic context does not decode the entire feature dictionary; it selects a small active set or low-dimensional subspace. The model below organizes that selection by sharing a token-associated subspace while placing sense-specific content in perpendicular residual subspaces. The global code size and this local organization are independent claims.

Fix one layer ℓ and one centered or whitened carrier for each test. The observed sequence context is x, while s denotes only the discrete operator class that indexes a candidate filter. In supervised experiments an operator class may be aligned with a labeled word sense; in unsupervised experiments it is latent. We abbreviate $h_\ell(t \mid x)$ by $h(x)$ when t and ℓ are fixed. The notation distinguishes mathematical types and roles:

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- P_t is the orthogonal projector onto $\mathcal{R}_t$;
- $S_{t,s} \subset \mathcal{R}_t^\perp$ is a low-dimensional sense subspace;
- $u_{t,s,j} \in S_{t,s}$ are vectors spanning that sense subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t, $u_{t,s,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,s}$ are subspaces. The w_i used in Sec. III have a separate role: they are generic feature directions in the geometric null model and are not reused as sense labels. This section resets the symbol x: it now denotes sequence context, not the random point on a sphere used in Sec. III A.

If $\mathcal{R}_t$ has an orthonormal basis collected in a matrix Q_t with p columns, then $\mathcal{R}_t^\perp = \ker(Q_t^\mathsf{T})$. Rank–nullity gives $\dim(\mathcal{R}_t^\perp) = d - p$. Thus every vector in a sense subspace is perpendicular to every reference direction; no new ambient dimension is being introduced.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \operatorname{span}\{r_t\}. \tag{27}$$

Empirically, r_t could be a normalized mean contextual embedding, a learned token-specific probe direction, or a static input embedding after an explicit map into the same layer carrier. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. This is the orthogonal projection theorem: for every chosen subspace, each vector has a unique closest component inside the subspace and a perpendicular residual. A weak "token component exists" test is also likely to pass simply because contextual states retain token identity. The substantive claim is that a subspace fixed from training data exposes additional held-out sense structure beyond token-mean subtraction, top-principal-component removal, and matched control subspaces.

For each operator class s, define a low-dimensional sense subspace of rank m_s

$$\begin{aligned} S_{t,s} &= \mathrm{span}\{u_{t,s,1}, \ldots, u_{t,s,m_s}\}, \\ S_{t,s} &\subset \mathcal{R}_t^{\perp}, \qquad m_s \ll d - \dim(\mathcal{R}_t). \end{aligned} \tag{28}$$

For the intended three-arm example, finance, river-edge, and seat/bench associations of "bank" correspond to different subspaces $S_{\mathrm{bank},s}$ inside the common orthogonal carrier $\mathcal{R}_{\mathrm{bank}}^{\perp}$. The verbal use in which an aircraft banks or tilts supplies a fourth sense and would add a fourth sense subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,s,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \mathrm{span}\{r_{t,1}, \ldots, r_{t,p}\}, \qquad p \geq 1, \tag{29}$$

and require

$$S_{t,s} \subset \mathcal{R}_t^{\perp}. \tag{30}$$

Equation (27) is the special case $p = 1$. Here p is the local rank of the reference subspace; it is not another global dimension. The same orthogonal-projection construction works for any declared p, which is why held-out comparison rather than existence carries the scientific burden. The bare subspace construction specifies only a pattern of sharing. In quantum logic, different contexts can share one or more observables or spectral projections; finite-dimensional examples with two shared observables are known [18]. In the semantic model, $\mathcal{R}_t$ is a common token-associated reference subspace, while the sense-specific subspaces lie in its orthogonal complement. These incidence relations alone imply neither noncommutativity nor quantum probability. The next subsection makes the stronger, optional operator structure explicit.

The low-rank assumption is empirical. Tying $\mathcal{R}_t$ across classes prevents each class model from independently relearning the proposed common component. This predicts steeper learning curves when examples per sense are scarce and a shared component is real; an unconstrained model should catch up with abundant data, and tying may hurt if the assumption is false. It also predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^{\perp}$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [19, 20].

B. Classical projectors and optional observable notation

The shared-subspace hypothesis can be parameterized by a constrained family of class-conditioned projectors. Let $Q_t \in \mathbb{R}^{d\times p}$ have orthonormal columns spanning $\mathcal{R}_t$, so that $P_t = Q_t Q_t^{\mathsf{T}}$. For each class s, let $B_{t,s} \in \mathbb{R}^{d\times m_s}$ have orthonormal columns spanning $S_{t,s}$. Here Q_t and $B_{t,s}$ are basis matrices, not additional semantic directions. The vector $Q_t^{\mathsf{T}} h$ contains the ordinary coordinates of h in the reference basis, and $Q_t Q_t^{\mathsf{T}} h$ reconstructs its orthogonal projection.

The projector onto the token-plus-sense subspace is

$$\Pi_{t,s} = P_t + B_{t,s} B_{t,s}^{\mathsf{T}}. \tag{31}$$

Because the two bases are orthonormal and perpendicular, their cross products vanish. Squaring $\Pi_{t,s}$ therefore returns $\Pi_{t,s}$ itself. This idempotence is the defining property of the orthogonal projector onto the direct sum $\mathcal{R}_t \oplus S_{t,s}$. All $\Pi_{t,s}$ have the common invariant sector $\mathcal{R}_t$ and differ on their sense sectors. In particular,

$$[\Pi_{t,s}, \Pi_{t,s'}] = [B_{t,s} B_{t,s}^{\mathsf{T}}, B_{t,s'} B_{t,s'}^{\mathsf{T}}] \tag{32}$$

need not vanish. The commutator is $[A, B] = AB - BA$. Expanding both products makes every term containing P_t vanish because the sense vectors lie in $\mathcal{R}_t^{\perp}$. A nonzero remainder means that applying two sense projections in opposite orders can give different vectors. Crucially, sharing a sector does not cause this order dependence. Moreover, two generic fitted projectors will often fail to commute, so a nonzero commutator alone is not evidence for a quantum-inspired account. It becomes discriminating only if its size predicts an independently specified order-sensitive contextual task or improves held-out prediction relative to matched projector models.

a. Optional observable notation. The real spectral theorem says that every symmetric matrix has an orthonormal eigenbasis. Borrowing quantum terminology, a symmetric class "observable" with the same basis-independent shared sector can be written

$$A_{t,s} = \lambda_t P_t + B_{t,s} \Lambda_{t,s} B_{t,s}^{\mathsf{T}}, \tag{33}$$

where the scalar λ_t is shared across classes and the diagonal $\Lambda_{t,s}$ is class-specific. In the orthonormal eigenbasis, $A_{t,s}$ assigns λ_t to every vector in $\mathcal{R}_t$, assigns the diagonal entries of $\Lambda_{t,s}$ to the chosen sense directions, and acts as zero on the displayed remainder. If P_t is to be the spectral projector for the shared eigenvalue, assume that λ_t is distinct from zero and from every diagonal entry of every $\Lambda_{t,s}$, as well as from the spectra of any added complementary terms. If $p > 1$, λ_t is deliberately degenerate within $\mathcal{R}_t$, so the observable identifies the shared subspace rather than an arbitrary basis within it. Additional orthogonal spectral terms may resolve that complement into further outcomes. The projector family is the empirical core of the proposal; the eigenvalues add scientific content only if they are prespecified or learned and improve a held-out observable outcome, rather than being assigned after the geometry is known.

The corresponding class-conditioned projection is concrete:

$$h_{t,s}^{(\Pi)}(x) = \Pi_{t,s} h(x) = P_t h(x) + B_{t,s} B_{t,s}^{\mathsf{T}} (I - P_t) h(x). \tag{34}$$

For $h(x) \neq 0$, make the observable operational through its normalized quadratic expectation and filtered state:

$$o_{t,s}(x) = \frac{h(x)^\mathsf{T} A_{t,s} h(x)}{\|h(x)\|^2}, \qquad h_{t,s}^{(A)}(x) = A_{t,s} h(x). \quad (35)$$

The first quantity is a Rayleigh quotient. In the eigenbasis supplied by the spectral theorem it is a weighted average of the eigenvalues, with weights equal to squared coordinates of $h(x)$. The filtered vector instead rescales each eigencomponent. Readers interested only in the classical projector model may omit $A_{t,s}$ and use $\Pi_{t,s}$ throughout. The router may use $o_{t,s}$ as a score, and a fitted module may use either $h_{t,s}^{(\Pi)}$ or $h_{t,s}^{(A)}$ as its class-conditioned representation. One may select a single s, or take a soft mixture of these outputs as in Sec. VI. Equations (31)– (35) define a proposed constrained module or probe; they do not claim that current LLMs already implement it. Quantum-logic discussions sometimes call contexts with shared outcomes *intertwining* or interlocking. Here "shared-sector" names that incidence pattern while avoiding confusion with the operator-theoretic assertion that an intertwiner X satisfies $AX = XB$. Everything here uses ordinary real matrices on classical hardware.

C. Projection energy and subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\mathrm{ref}}(x) + h_\perp(x), \\ h_{\mathrm{ref}}(x) &= P_t h(x), \\ h_\perp(x) &= (I - P_t) h(x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (36)$$

The orthogonal projection theorem makes this decomposition unique: $P_t h$ is the closest vector in $\mathcal{R}_t$ to h, and $(I - P_t)h$ is perpendicular to that subspace. Pythagoras gives $\|h\|^2 = \|P_t h\|^2 + \|(I - P_t)h\|^2$. The hypothesis is that most sense-disambiguating variation is present in $h_\perp(x)$, not that the decomposition itself is special. A basis-invariant score can be expressed directly with the orthonormal matrices already defined:

$$\begin{aligned} e_t(x) &= \|Q_t^\mathsf{T} h(x)\|^2 = \|P_t h(x)\|^2, \\ \beta_{t,s}(x) &= B_{t,s}^\mathsf{T} h_\perp(x), \\ f_{t,s}(x) &= \|\beta_{t,s}(x)\|^2 = \|B_{t,s} B_{t,s}^\mathsf{T} h_\perp(x)\|^2. \end{aligned} \quad (37)$$

The entries of $Q_t^\mathsf{T} h$ and $B_{t,s}^\mathsf{T} h_\perp$ are ordinary coordinates in orthonormal bases. Parseval's identity says that the sum of their squared coordinates equals the squared length of the corresponding orthogonal projection. This is why the energies do not depend on the particular orthonormal basis chosen inside the same subspace. Here e_t measures energy in the candidate reference subspace and

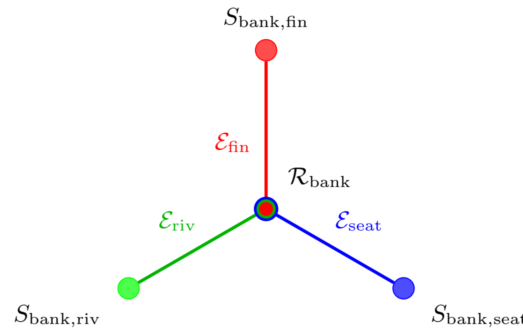

FIG. 1. Incidence schematic for the lexically ambiguous token "bank." The center is the single shared reference-subspace vertex $\mathcal{R}_{\mathrm{bank}}$ and the outer nodes are the financial, river-edge, and seat/bench sense subspaces in $\mathcal{R}_{\mathrm{bank}}^\perp$. Each colored arm denotes the displayed two-vertex fragment $\mathcal{E}_s = \{\mathcal{R}_{\mathrm{bank}}, S_{\mathrm{bank},s}\}$ associated with $\Pi_{\mathrm{bank},s}$; it is not a vector, map, or causal arrow. The concentric colors at the center mark membership of the same shared vertex in all three displayed context-hyperedge fragments. Additional outcome sectors would complete each measurement context in a full contextuality hypergraph.

$f_{t,s}$ measures projection energy in the candidate sense subspace. Both definitions work for any reference rank p and are invariant under changes of orthonormal basis within either subspace. An illustrative projection-energy score is

$$\gamma_{t,s}(x) = \frac{f_{t,s}(x)}{m_s}. \quad (38)$$

The division by the subspace rank matters when the m_s differ: for an isotropic residual, expected raw projection energy is proportional to rank. With equal ranks it does not change the ordering. A more careful empirical implementation may instead calibrate each raw energy against its own held-out null distribution. The selected sense label is

$$s^*(x) = \arg\max_s \gamma_{t,s}(x). \quad (39)$$

Multiplying every $\gamma_{t,s}$ by the same positive function of e_t would not change this ranking; reference energy can instead supply an independently calibrated engagement or abstention threshold. The score is an illustrative probe, not a derivation of how an LLM must disambiguate. At test time the true sense label is unavailable, so a practical implementation must learn the router from context without oracle access. The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than centered, token-mean, frequency-controlled, layer-matched, and random controls, the hypothesis fails.

Figure 1 illustrates these objects as an abstract hypergraph incidence structure.

D. Relation to dynamic contextual embeddings

Transformers already handle lexical ambiguity dynamically. The hidden state $h_\ell(t \mid x)$ of token t at layer ℓ depends on context x. The reference-subspace and shared-sector operator hypotheses are not alternative hardware architectures; they are proposed organizations and readouts of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t, write

$$\begin{aligned} h_\ell(t \mid x) &= P_t h_\ell(t \mid x) + z_\ell(t,x), \\ z_\ell(t,x) &= (I - P_t) h_\ell(t \mid x) \in \mathcal{R}_t^\perp. \end{aligned} \tag{40}$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t \mid x)$ tracks token identity or engagement, while $z_\ell(t,x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces. For each sense s, the projected states $z_\ell(t,x)$ should also concentrate near the low-dimensional subspace $S_{t,s} \subset \mathcal{R}_t^\perp$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace, and whether conditional application of $\Pi_{t,s}$ explains that movement better than matched unstructured projectors.

VI. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \qquad \alpha_j = \frac{\exp(q_{\mathrm{att}}^\mathsf{T} k_{\mathrm{att},j}/\sqrt{d_{\mathrm{head}}})}{\sum_{j'} \exp(q_{\mathrm{att}}^\mathsf{T} k_{\mathrm{att},j'}/\sqrt{d_{\mathrm{head}}})}. \tag{41}$$

$q_{\mathrm{att}}, k_{\mathrm{att},j} \in \mathbb{R}^{d_{\mathrm{head}}}$ are the query and key vectors of that head, and ν_j is its value vector. The local head width d_{head} is an architectural quantity, not necessarily the selected carrier dimension d used in the geometric null model. Keys determine the attention coefficients, while values are the vectors being averaged. Because they are produced by different learned maps, ordinary attention is routing rather than orthogonal projection or basis reconstruction.

Softmax makes every α_j nonnegative and makes the coefficients sum to one, so h' is a convex combination of the value vectors. Cauchy–Schwarz bounds a raw query–key score by the product of their lengths. The factor $1/\sqrt{d_{\mathrm{head}}}$ keeps typical score fluctuations of order one when many roughly independent coordinate contributions are added.

The useful analogy is therefore limited: attention amplifies some contextual directions and suppresses others. In the terminology of Sec. V, this resembles context-dependent subspace selection, but it is not the orthogonal projection used in the geometric model.

A usable context selector cannot receive the true sense label at inference. Let learned router logits $\ell_{\theta,s}(x,t)$, with router parameters θ, depend only on the available sequence and token state, and define

$$p_\theta(s \mid x,t) = \frac{\exp(\ell_{\theta,s}(x,t)/\tau_{\mathrm{r}})}{\sum_{s'} \exp(\ell_{\theta,s'}(x,t)/\tau_{\mathrm{r}})}, \qquad \tau_{\mathrm{r}} > 0. \tag{42}$$

Here τ_{r} is the router temperature: larger values flatten the class probabilities and smaller values sharpen them. The router may use the illustrative score in Eq. (38), but need not do so. Its weights define two possible shared-sector filters:

$$\begin{aligned} \bar{h}_t^{(\Pi)}(x) &= \sum_s p_\theta(s \mid x,t)\, \Pi_{t,s} h(x), \\ \bar{h}_t^{(A)}(x) &= \sum_s p_\theta(s \mid x,t)\, A_{t,s} h(x). \end{aligned} \tag{43}$$

Hard selection uses the largest router probability; soft selection uses Eq. (43). The effective operator $\sum_s p_\theta(s \mid x,t)\Pi_{t,s}$ is generally not idempotent and hence is not itself an orthogonal projector. Squaring a weighted sum creates cross terms, so the result need not equal the original sum. Attention may supply its routing signal, but neither the shared sector nor the operator structure follows from standard attention. The constrained model must therefore be compared with parameter-matched mixture-of-experts and unstructured low-rank alternatives.

For an implementable adapter at a chosen layer, let $Z_{t,s} \in \mathbb{R}^{d \times m_s}$ be a trainable seed matrix such that $(I - P_t)Z_{t,s}$ has full column rank, and set $B_{t,s} = \mathrm{orth}((I - P_t)Z_{t,s})$, where orth means ordinary Gram–Schmidt orthonormalization or an equivalent matrix factorization. Multiplication by $I - P_t$ first removes every component in the shared sector; full column rank ensures that m_s independent residual directions remain. Tie the same Q_t across all s, compute the router, and choose $\bar{h}_t \in \{\bar{h}_t^{(\Pi)}, \bar{h}_t^{(A)}\}$ from Eq. (43). The filtered state may be used as a diagnostic probe or inserted through the residual update $h^+ = h + \eta U \bar{h}_t$, where $U \in \mathbb{R}^{d \times d}$ is a learned output map and η is a scalar gate. The router, bases, spectra, and adapter are then trained on a language-model or task loss; parameter tying enforces the shared sector and the factor $(I - P_t)$ enforces transversality. Held-out evaluation must compare against adapters with the same ranks and parameter count but no shared-sector constraint.

VII. DISCUSSION

The chapter's positive geometric thesis is the abundance–access distinction. At a fixed angular tolerance, high dimension permits an exponentially large finite dictionary, but a decoder still pays for the features that activate together. The corresponding semantic hypothesis is concrete: training should allocate lower task-weighted cross-talk to important, frequently coactive features than appears in isotropic and activity-matched controls.

The shared-subspace proposal adds a separate inductive bias for lexical ambiguity. Reusing one token component across sense models should reduce sample complexity when the component is real and per-sense data are scarce. It does not add expressive power, and it should lose when forced sharing is wrong. Learning curves against equally expressive unstructured adapters or mixture models therefore test the proposed mechanism directly.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks may represent more features than they have neurons by using nonorthogonal directions with sparse activity [3, 21]. The null model separates dictionary coexistence from simultaneous linear readout. In the isotropic, weak-dependence model, many directions can coexist while a fixed-target correlation decoder accumulates interference of scale $\sqrt{k/d}$. Sparse activity controls that interference for this decoder; it is not asserted to be universally necessary for structured dictionaries or nonlinear recovery.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (17) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [22–25]. In the present framework, SAE decoder directions make the Lévy/JL calibration and coactivation-weighted diagnostic measurable, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. Relation to quantum cognition and compositional semantics

The phrase "quantum-inspired" has an established cognitive and linguistic lineage. Quantum-cognition models use Hilbert-space states, projectors, interference, and order effects as mathematical models of human judgment and concept combination, without requiring quantum processes in the brain [26, 27]. Work on the human mental lexicon has likewise used quantum structure to model word-association and memory experiments [28]. Those studies concern behavioral probabilities and conceptual judgments; the present chapter concerns hidden representations and learned readout in an LLM.

A different but neighboring tradition is compositional distributional semantics. Coecke, Sadrzadeh, and Clark combine vector-space word meanings with grammatical composition through tensor and categorical maps [29]. That construction shows how tensor products can organize composition, but it does not establish that a Transformer residual stream has a fully accessible semantic tensor factorization $d = q^n$.

Stripped of optional observable terminology, the model proposed here is a classical tied low-rank adapter: a router mixes sense-conditioned filters that share one subspace. That is its complete description unless sequential, noncommuting probes predict an order effect specified before the operators are fitted. Only such an additional result would make the operator language more than a convenient connection to the quantum-cognition literature.

C. Limitations

The limitations can be grouped into four issues.

Geometry. Lévy's lemma controls a fixed Lipschitz probe; a finite dictionary additionally needs a union bound or code construction. JL gives an existence scale, the random-code formula gives an ensemble calibration, and the Welch bound gives a deterministic floor. None is a semantic upper bound.

Measurement protocol. The usable rank d and dictionary size M depend on layer choice, whitening, feature extraction, normalization, and duplicate removal. Contextual states can be anisotropic [30–33], although anisotropy is not universal [34]. Whitening changes the inner product and therefore changes angles. Finite-dimensional norm equivalence preserves convergence and continuity, but it does not preserve numerical angles or coherence. Every comparison must therefore use one declared metric and selection rule.

Readout. The law $\sigma_{\text{int}} \sim \sqrt{k/d}$ concerns one correlation decoder under random-like cancellation. Structured dictionaries, nonlinear decoders, inactive-feature comparisons, and correlated errors can change it. Likewise, Eq. (24) omits cross-moment terms unless the stated zero-cross-moment assumption holds.

Shared-subspace model. The model may simply be wrong. Token identity can make its weak version pass while its nontrivial held-out comparison fails. Ranks, references, routers, and optional spectra must be fixed on training data; the true sense label is unavailable at test time; and a nonzero fitted commutator is generic. The proposed controls decide whether the shared model adds held-out value beyond these simpler explanations.

D. Proposed experiments and failure criteria

No experiment is reported in this chapter. The purpose of this section is to turn the analytical claims into a staged, falsifiable program. The first stage calibrates geometry; the later stages test increasingly restrictive semantic structure.

Geometric calibration (not a semantic claim). For a dictionary independently established to be overcomplete, compare its coherence, overlap distribution, and fixed-target readout error with isotropic and activity-matched controls. Agreement with Eq. (12) is a baseline result, not evidence that overcompleteness is useful; departure from it must be diagnosed before being called learned structure. JL itself is a theorem; the empirical question is whether task-relevant geometry and readout survive projection near its logarithmic dimension scale.

Claim A0: token-component sanity check. A token-associated component explains cross-context variation shared by occurrences of the same token. This weak result is expected and is not evidence for a privileged semantic sector.

Claim A1: nontrivial reference-subspace claim. Relative to token-mean subtraction, top-principal-component removal, and same-rank random or supervised controls, a reference subspace $\mathcal{R}_t$ fixed from training data improves held-out sense prediction, compression, or additional predictive information in $\mathcal{R}_t^\perp$.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^\perp$, sense-conditioned states admit a cross-validated low-rank description that predicts or compresses held-out states better than shuffled-label, ordinary clustering, and matched-subspace controls. Related polysemous senses need not have minimal cross-sense coherence.

Claim C: sample-efficient shared-sector filters. A non-oracular router and class-conditioned projectors $\Pi_{t,s}$ sharing P_t improve held-out contextual prediction or compression relative to parameter-matched unstructured projectors and mixture-of-experts models, with the largest gain when per-sense data are scarce or imbalanced. A commutator matters only if it predicts order effects defined independently of the fitted operators.

These claims can fail independently. The geometric calibration can hold while all structural claims fail. A0 may pass while A1 fails; that outcome would reject a privileged reference subspace without denying ordinary token identity. Claim B can hold relative to a non-token-associated subspace, and A1 and B can hold even if the tying in Claim C adds no predictive value.

a. Minimum pilot. A small first study needs no retraining of the language model. Choose one public model, one layer, and a prespecified sample of ambiguous tokens from the Word-in-Context benchmark or SemCor. Estimate candidate token means and top principal components on the training split only. Train the same simple linear sense classifier on the full states, on residuals after each candidate removal, and on same-rank random controls. Held-out accuracy and calibration decide A1. Adding sparse-autoencoder features then tests the coactivation statistic. A failure at this stage should stop stronger operator claims rather than be hidden by a more flexible router.

The following tests separate the possibilities.

1. *Lévy/random-code calibration.* Fix the carrier metric, spectral threshold or whitening rule, feature extractor, and duplicate criterion before evaluation. Report the selected rank d, dictionary size M, maximum coherence, the full overlap distribution, and active-set statistics. Compare with the Welch floor, isotropic dictionaries, and controls matched for marginal activity and coactivation. Treat Eq. (12) as a finite-sample null prediction, not a pass/fail capacity test. Also compare the signed pairwise distribution with the independently isotropic variance $1/d$; deviations may reflect dependence, anisotropy, extraction choices, or learned alignment.

2. *JL projection-stability test.* Fix a finite evaluation set of M normalized directions or contextual states, include the origin, and prespecify a relative squared-distance distortion δ_{JL}. Project to d_{proj} dimensions using frozen random maps scaled to preserve squared lengths on average, and measure pairwise distortion and held-out task performance as d_{proj} varies. JL predicts metric preservation when d_{proj} is proportional to $\delta_{\mathrm{JL}}^{-2}\log(M+1)$ up to universal constants; preservation of semantic performance is the additional empirical question. Compare random Gaussian or orthogonal maps with learned low-rank maps at the same d_{proj}.

3. *Reference-subspace test.* For a lexically ambiguous token, choose candidate reference subspaces generated by the mean contextual embedding, a static input embedding explicitly mapped into the same layer carrier, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (40). Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements obtained after ordinary centering, token-mean subtraction, top-principal-component removal, and random, frequency-matched, or layer-matched control subspaces of the same rank. Compare held-out classification, compression, or incremental mutual information, with permutation over sense labels and bootstrap over token instances.

4. *Operational subspace test.* For each sense s, collect projected states $z_\ell(t,x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by principal component analysis (PCA) or probabilistic factor analysis. PCA is an application of the spectral theorem to the sample covariance matrix: its directions are eigenvectors with the largest eigenvalues. Select m_s without test-label leakage and use an orthonormal basis $B_{t,s}$. Evaluate held-out reconstruction and sense prediction against shuffled labels, ordinary supervised subspaces, and unsupervised clustering of equal rank. Report $\|B_{t,s}^\mathsf{T} B_{t,s'}\|_{\mathrm{op}}$ descriptively, without assuming related senses must be nearly orthogonal.

5. *Reference-versus-sense readout test.* Linear probes trained on $z_\ell(t,x)$ should perform comparably

to probes trained on the full state $h_\ell(t \mid x)$, while probes restricted to the reference component $P_t h_\ell(t \mid x)$ should provide less incremental sense information after frequency, syntax, and sense-prior controls. Near-chance performance is not required because the reference component may legitimately retain correlated information.

6. *Shared-sector operator test.* Fit P_t, $B_{t,s}$, $A_{t,s}$, and the router on training sequences, then evaluate Eq. (43) without true sense labels on held-out instances. Compare with unconstrained low-rank projectors, ordinary contextual probes, parameter-matched mixture-of-experts models, and shuffled class or sense labels. If an order claim is made, define the sequential linguistic or model intervention independently; then ask whether $\|[\Pi_{t,s}, \Pi_{t,s'}]\|_{\mathrm{op}}$ predicts the measured asymmetry. Merely obtaining a nonzero commutator does not pass the test. Repeat the comparison across training-set sizes and sense-imbalance levels. The tying mechanism predicts a larger advantage in the low-data regime and no guaranteed advantage when data are abundant or the shared component is weak.

7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions

$$h_k = h_0 + \lambda \sum_{j=1}^{k} x_j w_j, \tag{44}$$

where h_0 is a held-out baseline activation, the w_j are independently chosen unit injection directions, the coefficients x_j are independent, centered, and unit-variance, and λ controls injection amplitude. The directions may come, for example, from SAE decoder directions or frozen linear probes. For random-like directions under the specified correlation decoder, the injected fixed-target interference has RMS scale $\lambda\sqrt{k/d}$, up to the distinction between k and $k-1$. How that interference changes downstream task performance is an empirical question. Include norm-preserving, random-direction, activation-frequency, and out-of-distribution controls; degradation by itself does not validate semantic superposition.

8. *Coactivation-dependent geometry.* At matched marginal activation frequency, feature similarity, SAE reconstruction error, amplitude, and task relevance, important frequently coactive features should contribute less to Eq. (24) than activity-matched controls. This tests learned redistribution of interference, independently of the reference-subspace hypothesis.

E. Scope of the quantum analogy

The complete implementation described in this chapter is classical: a router mixes real low-rank filters that share a real subspace. The connection to quantum contextuality is structural. Each $\Pi_{t,s}$ acts identically on $\mathcal{R}_t$, while the relative geometry of the residual sense subspaces can make two projections order-dependent. Figure 1 records that shared-incidence pattern. The reference rank is explicit: $p = 1$ shares one direction, and $p > 1$ shares a subspace of p directions.

There are two progressively stronger uses of quantum language. First, projectors and symmetric operators can serve as an abstract contextual model on a classical computer. This use becomes scientifically distinctive only if the constraints improve held-out prediction or if sequential operators predict an order effect defined before fitting. Second, a genuinely quantum machine would encode states in physical qubits or other quantum degrees of freedom, use phase-sensitive unitary evolution, and apply Born probabilities at measurement. A standard Transformer and the proposed adapter do none of these things.

Quantum contextuality in physics has the force of no-go theorems such as Kochen–Specker [35, 36]. No analogous theorem is claimed for semantic embeddings. Generic real projectors can fail to commute, so a nonzero fitted commutator alone supplies no evidence.

VIII. CONCLUSION

The central organizing distinction is between abundance and access. Lévy concentration, the Johnson–Lindenstrauss lemma, and random spherical coding are related expressions of the same high-dimensional concentration phenomenon: at fixed angular tolerance, a d-dimensional carrier admits exponentially many directions. The Welch bound gives a deterministic floor on their worst overlap. None of these counts makes the directions simultaneously readable. A simple readout from k random-like active features instead pays an RMS interference cost of order $\sqrt{k/d}$.

This distinction yields the chapter's positive semantic prediction. A useful learned dictionary should not merely contain many directions; it should place lower task-weighted cross-talk between important features that often coactivate. Equation (24) turns that idea into a held-out comparison against isotropic and activity-matched controls.

For lexical ambiguity, the proposed tied adapter reuses a token-associated subspace and learns low-rank residual sense subspaces. Sharing is predicted to improve sample efficiency when a genuine common component exists and per-sense data are scarce. The weak existence of a token component is trivial; the substantive test is incremental held-out value beyond token means, principal-component removal, and parameter-matched unstructured adapters.

The minimum pilot in Sec. VII D can reject the reference-subspace claim before the optional operator layer is attempted. The implementation is classical. It becomes specifically quantum-like only if sequential operator constraints predict prespecified order effects beyond classical baselines. The doubly exponential tensor-product count remains only the conditional remark of Sec. III C, not a conclusion about ordinary LLMs.

Appendix A: Explicit formalization of the three computational loops

The following subsections correspond one-to-one to Secs. II A–II C. The purpose is not to specify every engineering detail of a modern model. It is to identify, in the order in which they occur, which objects are discrete symbols, which are vectors, which are temporary results of a calculation, and which are persistent model parameters. No argument about the semantic geometry in the main text depends on this appendix.

We first fix the objects used throughout. Let $\mathcal{S}$ be the set of finite character strings and let $\mathcal{V} = \{1, \dots, V\}$ be a finite vocabulary of V token identifiers. A token identifier is an integer label; it is not itself a vector. For a nonnegative integer T, the Cartesian power $\mathcal{V}^T$ is the set of token sequences of length T; the union below allows every finite length. The tokenizer is a fixed, variable-length map

$$\mathsf{Tok} : \mathcal{S} \longrightarrow \bigcup_{T \geq 0} \mathcal{V}^T. \tag{A1}$$

Thus Tok turns one string into a finite sequence of identifiers. Its companion Detok assembles a token sequence into text. Apart from any declared normalization of spaces or characters, detokenization reverses the tokenization of an admissible string. Neither map learns a contextual meaning during the Transformer calculation.

The symbol d_{model} denotes the architectural width of the residual stream, so every hidden state below is an ordinary real column vector in $\mathbb{R}^{d_{\text{model}}}$. It must not be confused with the usable semantic carrier rank d in the main text: restriction, whitening, or removal of inactive directions may give $d \leq d_{\text{model}}$, with equality when the full residual stream is used. The symbol L denotes the number of Transformer layers. The symbol Θ denotes the complete collection of persistent numerical parameters: the token embedding table, the matrices in all attention and feed-forward blocks, normalization parameters, and the final vocabulary readout. If all their entries are written in one long list, then Θ may be regarded as a vector in $\mathbb{R}^{N_{\text{par}}}$, where N_{par} is the number of trainable scalar parameters. This flattening is only bookkeeping; an implementation stores the parameters as many arrays of different shapes. Thus d_{model}, V, and N_{par} count three different things: architectural hidden coordinates, token types, and scalar model parameters.

Persistent parameters survive from one prompt to the next. Hidden states, query, key, and value vectors, attention coefficients, and next-token probabilities are transient quantities recomputed for a particular input. The phrase "attention weight" is therefore potentially confusing: an attention coefficient is a temporary number, whereas a model weight is an entry of a persistent matrix in Θ. A superscript in parentheses denotes a stage or index rather than exponentiation: $h_i^{(\ell)}$ will index a layer, while $\Theta^{(r)}$ with $r = 0, 1, 2, \dots$ will index a successive parameter update.

1. Creating the weights

This subsection follows Sec. II A: text is tokenized, tokens are represented by vectors, self-attention makes those vectors context-dependent, a next-token error is calculated, and an optimizer changes the persistent parameters.

a. Step 1: turn text into discrete identifiers. Consider one training string $\sigma \in \mathcal{S}$ whose selected training segment contains $T \geq 2$ tokens, so that it supplies at least one next-token prediction. Systems often use declared beginning- or end-of-sequence tokens at document boundaries. For $i \in \{1, \dots, T\}$, let $x_i \in \mathcal{V}$ denote the identifier at position i. Tokenization gives

$$(x_1, \dots, x_T) = \mathsf{Tok}(\sigma) \in \mathcal{V}^T. \tag{A2}$$

The same identifier may occur several times. At this stage all of its occurrences carry the same discrete label, irrespective of context.

b. Step 2: look up an initial vector and mark its position. Let $E_{\text{tok}} \in \mathbb{R}^{V \times d_{\text{model}}}$ be the token embedding table. It is one of the trainable arrays contained in Θ. Its row $E_{\text{tok}}[a] \in \mathbb{R}^{d_{\text{model}}}$ is the lookup vector for token type $a \in \mathcal{V}$; square brackets mean table lookup. In the simplest additive description, let $\rho_i^{\text{pos}} \in \mathbb{R}^{d_{\text{model}}}$ be a position vector. The initial hidden vector $h_i^{(0)} \in \mathbb{R}^{d_{\text{model}}}$ is

$$h_i^{(0)} = E_{\text{tok}}[x_i] + \rho_i^{\text{pos}}, \qquad i = 1, \dots, T. \tag{A3}$$

The superscript (0) means "before the first Transformer layer." The position vector may be fixed or learned; when it is learned, its entries also belong to Θ. Systems using rotary or other positional schemes do not literally add ρ_i^{pos}, but they serve the same present purpose: the calculation must know both which token occurred and where it occurred. Two occurrences of "bank" begin with the same row of E_{tok}, but their later hidden states can differ because their surrounding tokens and positions differ.

c. Step 3: make the vectors context dependent by self-attention. We spell out one attention head in one layer. Let $h_i^{(\ell-1)} \in \mathbb{R}^{d_{\text{model}}}$ be the vector at position i entering layer $\ell \in \{1, \dots, L\}$. Let d_{head} be the width of one head, and let $W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)} \in \mathbb{R}^{d_{\text{head}} \times d_{\text{model}}}$ be three learned matrices contained in Θ. For target position i and source position j, write q_i, k_j, v_j for the three outputs

$$\begin{aligned} q_i &= W_Q^{(\ell)} h_i^{(\ell-1)}, \\ k_j &= W_K^{(\ell)} h_j^{(\ell-1)}, \\ v_j &= W_V^{(\ell)} h_j^{(\ell-1)}. \end{aligned} \tag{A4}$$

The vectors $q_i, k_j, v_j \in \mathbb{R}^{d_{\text{head}}}$ are called the query, key, and value. A query describes what information position i is seeking, a key describes how position j may be matched, and a value contains the information to be combined if that position receives attention. These phrases interpret three ordinary matrix–vector products; they do not introduce new kinds of mathematical objects.

For a decoder-only model, position i may attend only to positions $j \le i$. Let s_{ij} denote the real comparison score, let α_{ij} denote the normalized attention coefficient, and let o_i^{att} denote the output of this head. The letter t below is only a dummy source-position index. These quantities are

$$\begin{aligned} s_{ij} &= \frac{q_i^{\mathsf{T}} k_j}{\sqrt{d_{\text{head}}}}, \\ \alpha_{ij} &= \frac{\exp(s_{ij})}{\sum_{t=1}^{i} \exp(s_{it})}, \\ o_i^{\text{att}} &= \sum_{j=1}^{i} \alpha_{ij} v_j. \end{aligned} \tag{A5}$$

The first line uses the Euclidean inner product. By the Cauchy–Schwarz inequality, $|q_i^{\mathsf{T}} k_j| \le \|q_i\| \|k_j\|$, so a large comparison requires suitably aligned query and key vectors as well as suitable lengths. The factor $1/\sqrt{d_{\text{head}}}$ controls the numerical scale of the scores. The ordinary exponential function is positive, hence every $\alpha_{ij} \ge 0$, and direct summation gives $\sum_{j=1}^{i} \alpha_{ij} = 1$. Consequently o_i^{att}, the output of this attention head, is a weighted average, or convex combination, of the permitted value vectors. The restriction $j \le i$ is the causal mask: it prevents the calculation at position i from using the future target x_{i+1}.

A practical layer has several heads. Their outputs are concatenated, mapped back to $\mathbb{R}^{d_{\text{model}}}$, combined with a residual connection, normalized, and passed through a feed-forward block. Denote the resulting vector by $h_i^{(\ell)} \in \mathbb{R}^{d_{\text{model}}}$. Repeating the procedure for $\ell = 1, \ldots, L$ yields final contextual vectors $h_i^{(L)}$. Let F_Θ denote the complete layered Transformer map. Then

$$(h_1^{(L)}, \ldots, h_T^{(L)}) = F_\Theta(h_1^{(0)}, \ldots, h_T^{(0)}) \tag{A6}$$

Equation (A6) collects all these operations into one function. During this forward pass Θ is fixed. In contrast, $q_i, k_j, v_j, \alpha_{ij}$, and $h_i^{(\ell)}$ depend on the current string and are transient.

d. Step 4: turn the final vector into a next-token distribution. Let $W_{\text{out}} \in \mathbb{R}^{V \times d_{\text{model}}}$ and $b \in \mathbb{R}^V$ be the vocabulary readout matrix and bias; both belong to Θ. At training position i, let $g_i \in \mathbb{R}^V$ denote the score vector. For a candidate token $a \in \mathcal{V}$, let $p_\Theta(a \mid x_1, \ldots, x_i)$ denote its predicted probability given the prefix. The letter c below is only a dummy token index in the denominator. Define

$$\begin{aligned} g_i &= W_{\text{out}} h_i^{(L)} + b, \\ p_\Theta(a \mid x_1, \ldots, x_i) &= \frac{\exp(g_{i,a})}{\sum_{c \in \mathcal{V}} \exp(g_{i,c})}. \end{aligned} \tag{A7}$$

The entry $g_{i,a}$ is a real-valued score, often called a logit, not a probability. The denominator is a finite sum of positive numbers, so the softmax expression is nonnegative and sums to one over $a \in \mathcal{V}$. The vertical bar means "given the prefix $(x_1, \ldots, x_i)$." Because of the causal mask, this distribution depends on that prefix and not on the unseen suffix $(x_{i+1}, \ldots, x_T)$.

e. Step 5: compare the prediction with the observed continuation. Let $P_\Theta(x_2, \ldots, x_T \mid x_1)$ denote the sequence-level conditional probability assigned to the observed continuation. The multiplication rule for conditional probabilities gives

$$P_\Theta(x_2, \ldots, x_T \mid x_1) = \prod_{i=1}^{T-1} p_\Theta(x_{i+1} \mid x_1, \ldots, x_i). \tag{A8}$$

This product does not assume that tokens are independent: every factor is conditioned on the entire available prefix. Taking the negative natural logarithm turns the product into a sum. The loss for this string is

$$\mathcal{L}_\sigma(\Theta) = -\sum_{i=1}^{T-1} \log p_\Theta(x_{i+1} \mid x_1, \ldots, x_i). \tag{A9}$$

Each summand is small when the observed next token was assigned a probability near one and large when it was assigned a probability near zero.

f. Optional calculus detail: the initial error signal. The following derivative may be skipped without losing the computational sequence. Let ℓ_i be the ith summand in Eq. (A9), and let the indicator $\mathbf{1}_{\{a=x_{i+1}\}}$ equal one when a is the observed next token and zero otherwise. Differentiating the logarithm and softmax gives

$$\frac{\partial \ell_i}{\partial g_{i,a}} = p_\Theta(a \mid x_1, \ldots, x_i) - \mathbf{1}_{\{a=x_{i+1}\}}. \tag{A10}$$

Thus the derivative compares the predicted probability with a target that places all its mass on the observed token. This is the initial error signal propagated backward through the model.

g. The parameter update. Let $\mathcal{B}$ be a finite minibatch of training strings and let $\widehat{\mathcal{L}}_{\mathcal{B}}$ be the average of their string losses, usually normalized by the number of predicted tokens. The hat emphasizes that this is a finite-sample average. Let G denote its gradient with respect to all parameters, let $\eta > 0$ be a learning rate, and let

Θ_{new} denote the parameters after one update. Then

$$G = \nabla_\Theta \widehat{\mathcal{L}}_{\mathcal{B}}(\Theta), \qquad \Theta_{\mathrm{new}} = \Theta - \eta G, \qquad \eta > 0. \tag{A11}$$

After flattening Θ into $\mathbb{R}^{N_{\mathrm{par}}}$, the gradient ∇_Θ is the vector of partial derivatives with respect to every parameter, so G has the same number of scalar entries as Θ. Backpropagation computes it by repeatedly applying the multivariable chain rule through the readout, feed-forward blocks, and attention operations.

h. Optional calculus detail: why subtract the gradient? The symbol $\|G\|$ denotes the Euclidean length of G, and $\approx$ below means equality to first order in the small step size. The elementary first-order Taylor formula explains the minus sign: for a sufficiently small step and a differentiable loss,

$$\widehat{\mathcal{L}}_{\mathcal{B}}(\Theta - \eta G) \approx \widehat{\mathcal{L}}_{\mathcal{B}}(\Theta) - \eta \|G\|^2. \tag{A12}$$

The omitted remainder becomes negligible compared with the linear term as $\eta \to 0$. Therefore a nonzero negative-gradient step locally lowers the same batch loss when the step is small enough; it does not promise a global minimum. Practical optimizers use running averages, adaptive scaling, regularization, and many batches, so they need not equal this simple update. The logical division remains: self-attention changes temporary activations, whereas the optimizer changes persistent parameters. Starting from initialized values and repeating the forward–loss–backward–update cycle produces the trained weights.

All spaces in this derivation are finite-dimensional Euclidean spaces. Calling $\mathbb{R}^{d_{\mathrm{model}}}$ a real Hilbert space is formally correct because it is complete, but no completeness argument and no quantum postulate is needed. The linear algebra consists of vectors, matrices, inner products, and finite sums; the calculus consists mainly of differentiation and the chain rule.

2. Generating the output

This subsection follows Sec. II B. The tokenizer and forward calculation are the same as during training, but the observed next token is now unavailable. The model must choose one token, append it, and use the enlarged prefix for the next choice.

a. Initial state of the loop. Let $\sigma_{\mathrm{prompt}} \in \mathcal{S}$ be the complete prompt, including any visible role or control tokens. Suppose tokenization returns m identifiers, where m is the prompt length, and let $u_i \in \mathcal{V}$ be the identifier at prompt position i. Then

$$(u_1, \ldots, u_m) = \mathsf{Tok}(\sigma_{\mathrm{prompt}}) \in \mathcal{V}^m. \tag{A13}$$

Each u_i is an integer token identifier, not a vector. We use u rather than the training symbol x only to keep the two discussions visually separate. Let $\Theta^\star$ denote the deployed parameter values obtained after training and any declared post-training. The star labels the deployed checkpoint; it does not denote complex conjugation. Ordinary generation holds $\Theta^\star$ fixed.

Apply the lookup and Transformer operations in Eqs. (A3)–(A6) to the prompt. After the prefix has length $j \geq m$, let $h_j^{(L)}$ be its last contextual vector and let $g_j = W_{\mathrm{out}} h_j^{(L)} + b \in \mathbb{R}^V$ be the score vector. Its entry $g_{j,a}$ is the score for candidate token $a \in \mathcal{V}$.

b. Convert the scores into a decoding rule. Choose a positive decoding temperature τ_{dec}. Let $\pi_j(a)$ denote the normalized probability of candidate token a, and let $u_{1:j}$ abbreviate the prefix $(u_1, \ldots, u_j)$. The index c below is a dummy token index. The notation $u_{j+1} \sim \pi_j$ will mean that the next token identifier is sampled from π_j; here $\sim$ means sampling, not approximate equality. Define

$$\begin{aligned} \pi_j(a) &= \frac{\exp(g_{j,a}/\tau_{\mathrm{dec}})}{\sum_{c\in\mathcal{V}} \exp(g_{j,c}/\tau_{\mathrm{dec}})}, \qquad a \in \mathcal{V}, \\ u_{j+1} &\sim \pi_j, \\ u_{1:j+1} &= (u_{1:j}, u_{j+1}). \end{aligned} \tag{A14}$$

The parentheses in the last line mean sequence concatenation. The function $\pi_j : \mathcal{V} \to [0,1]$ is a probability distribution because its entries are nonnegative and its denominator makes $\sum_{a\in\mathcal{V}} \pi_j(a) = 1$.

Thus a token identifier is not being equated with a probability. Score differences determine probability ratios:

$$\frac{\pi_j(a)}{\pi_j(c)} = \exp\left(\frac{g_{j,a} - g_{j,c}}{\tau_{\mathrm{dec}}}\right). \tag{A15}$$

A smaller positive temperature magnifies score differences and makes the distribution more concentrated; a larger temperature flattens it. In greedy decoding, sampling is replaced by $u_{j+1} = \arg\max_{a\in\mathcal{V}} g_{j,a}$, meaning that the candidate token with the largest score is selected. Other rules may retain only a declared subset of candidates and then renormalize. These are decoding choices, not additional training steps.

c. Feed the chosen token back as part of the next input. The last line of Eq. (A14) appends u_{j+1} to the prefix. The enlarged sequence is then passed through the same fixed-parameter calculation to obtain g_{j+1} and the following distribution. This is a recurrence: the state used at one step contains the output of the preceding step. It is not gradient descent because no derivative is taken and $\Theta^\star$ does not change. A token selected early in the response can alter all later query–key comparisons and therefore all later distributions.

Let J be the final sequence position and let $u_{m+1:J}$ abbreviate the complete generated continuation $(u_{m+1}, \ldots, u_J)$. Write $\mathbb{P}_{\Theta^\star,\tau_{\mathrm{dec}}}$ for the probability assigned by the fixed deployed model and the displayed temperature-sampling rule. The multiplication rule gives

$$\mathbb{P}_{\Theta^\star,\tau_{\mathrm{dec}}}(u_{m+1:J} \mid u_{1:m}) = \prod_{j=m}^{J-1} \pi_j(u_{j+1}). \tag{A16}$$

This product does not assert independence. Each factor π_j was calculated from the complete preceding prefix $u_{1:j}$. Generation may stop when a declared end token is chosen, a length limit is reached, or another stopping rule fires. The newly generated answer is

$$\sigma_{\text{answer}} = \mathsf{Detok}(u_{m+1}, \ldots, u_J). \tag{A17}$$

The prompt tokens are omitted because this expression denotes only the new answer text.

An implementation usually stores previously computed key and value vectors in a key–value cache. The cache avoids recomputing the same transient quantities for the unchanged prefix. It changes time and memory use, but it is neither an external factual memory nor a change to the parameters $\Theta^\star$. In summary, the token sequence, attention coefficients, and hidden vectors change at every generation step; the deployed model weights do not.

3. Recursive adaptation, alignment, and hallucination

This subsection expands Sec. II C. The word "recursive" can refer to three different loops, and only the third changes model weights.

a. Loop 1: autoregressive state recursion. Let $c_j = u_{1:j}$ denote the token prefix at generation step j. The update $c_{j+1} = (c_j, u_{j+1})$ changes the input state, while the deployed parameter vector remains $\Theta^\star$ at every token step. This is precisely the generation loop in Eq. (A14). It supplies new context without learning new persistent parameters.

b. Loop 2: external memory or retrieval. Let $r = 0, 1, 2, \ldots$ count complete interaction rounds, let I_r and O_r be the user's input and the model's output in round r, and let $\mathcal{M}$ be the set of permitted external-memory records. The symbol $m_r \in \mathcal{M}$ denotes the record after round r and is unrelated to the prompt-length integer m in the preceding subsection. Let U_{mem} be a rule that revises the record, let A_{mem} be a rule that assembles retrieved material into a prompt, and let $\sigma^{(r+1)}_{\text{prompt}}$ denote the resulting prompt for the next round. A memory system may use maps of the schematic form

$$\begin{aligned} m_{r+1} &= U_{\text{mem}}(m_r, I_r, O_r), \\ \sigma^{(r+1)}_{\text{prompt}} &= A_{\text{mem}}(I_{r+1}, m_{r+1}). \end{aligned} \tag{A18}$$

The first line updates the external record; the second uses that record when constructing the next input string. These maps may be implemented by a database, a retrieval system, or a longer conversation history; they need not be linear. They alter what the model reads but still leave $\Theta^\star$ fixed. This is why a key–value cache, a retrieved document, and a trained parameter are three different kinds of storage.

c. Loop 3: genuine online parameter adaptation. Let $\Theta^{(r)} \in \mathbb{R}^{N_{\text{par}}}$ denote the parameter vector available at the beginning of interaction round r. The starting value $\Theta^{(0)}$ may already include instruction or preference-based post-training. Let $\mathcal{B}_r$ be a finite collection of externally verified input–target examples approved for adaptation after round r, and let $\mathcal{J}_r(\Theta; \mathcal{B}_r)$ be the scalar loss chosen for those examples. Let $g_r \in \mathbb{R}^{N_{\text{par}}}$ denote the gradient of that loss at the current parameters. Thus

$$g_r = \nabla_\Theta \mathcal{J}_r(\Theta^{(r)}; \mathcal{B}_r) \in \mathbb{R}^{N_{\text{par}}}. \tag{A19}$$

Thus g_r lists the derivative of the adaptation loss with respect to every scalar model parameter.

To state which parameters may change, let $\mathcal{P}^{(\Theta)}_r : \mathbb{R}^{N_{\text{par}}} \to \mathbb{R}^{N_{\text{par}}}$ be a declared linear mask or projection. Choose a learning rate $\eta_r > 0$, and let $\Theta^{(r+1)}$ denote the parameter vector after the update. A restricted gradient step is

$$\Theta^{(r+1)} = \Theta^{(r)} - \eta_r \mathcal{P}^{(\Theta)}_r g_r, \qquad \eta_r > 0. \tag{A20}$$

In words: the new parameters equal the old parameters minus a chosen step size times the permitted part of the adaptation gradient.

If $\mathcal{P}^{(\Theta)}_r = 0$, no parameter changes. If it keeps only adapter coordinates, only that adapter changes. If it is the identity map, every parameter direction is permitted.

d. Optional linear-algebra detail: restricted descent. When $\mathcal{P}^{(\Theta)}_r$ is an orthogonal projection, it satisfies $(\mathcal{P}^{(\Theta)}_r)^\mathsf{T} = \mathcal{P}^{(\Theta)}_r$ and $(\mathcal{P}^{(\Theta)}_r)^2 = \mathcal{P}^{(\Theta)}_r$. Elementary linear algebra then splits g_r into an allowed part $\mathcal{P}^{(\Theta)}_r g_r$ and a perpendicular forbidden part. For a sufficiently small step, the same first-order argument as in Eq. (A12) gives

$$\mathcal{J}_r(\Theta^{(r+1)}; \mathcal{B}_r) \approx \mathcal{J}_r(\Theta^{(r)}; \mathcal{B}_r) - \eta_r \|\mathcal{P}^{(\Theta)}_r g_r\|^2. \tag{A21}$$

The equality $g_r^\mathsf{T} \mathcal{P}^{(\Theta)}_r g_r = \|\mathcal{P}^{(\Theta)}_r g_r\|^2$ follows from symmetry and idempotence. If $\mathcal{P}^{(\Theta)}_r g_r \neq 0$, this calculation explains strict local descent on the declared adaptation loss for a sufficiently small step. It does not guarantee preservation of earlier abilities.

The superscript (Θ) reminds us that this operation takes place in the large parameter space. It is unrelated to the token-space projector P_t in Sec. V. Equation (A20) restricts an update direction but does not by itself bound the update size. Clipping, a trust region, or an explicit norm bound can limit one step. A general projected-gradient algorithm may additionally project the updated point back onto a feasible set; that is a further operation not displayed here.

A user interaction normally supplies no verified target. Unrestricted updates can cause forgetting, distribution drift, privacy leakage, or data poisoning, and repeated training on unverified model output can reinforce errors [37]. A defensible online learner therefore needs provenance, external verification, bounded or adapter-local updates, held-out tests, and a recoverable reference model.

e. Alignment is a separate parameter intervention. Let $\Theta_{\rm base}$ denote a fixed base model before a declared alignment procedure, and let $\mathcal{D}_{\rm align}$ be the data used by that procedure. Let Align denote the complete post-training procedure and let $\Theta_{\rm align}$ denote its resulting parameters. Schematically,

$$\Theta_{\rm align} = \mathsf{Align}(\Theta_{\rm base}; \mathcal{D}_{\rm align}). \tag{A22}$$

The procedure need not be a linear map. Alignment therefore changes persistent weights, but it usually occurs before deployment rather than after every user interaction. The difference $\Theta_{\rm align} - \Theta_{\rm base}$ records a parameter change; its existence or length alone does not determine whether factual accuracy improves or worsens. That direction of effect must be measured against a matched control.

f. Define the measurement before comparing models. Let q denote one evaluation case, including its prompt and any grounding material supplied to the model, and let $\mathcal{D}_{\rm eval}$ be the fixed distribution of such cases. Let Y denote the generated response under one fixed decoding rule. Finally, let $Z(q,Y) \in \{0,1\}$ be a prespecified response-level indicator: it is one if the response contains an unsupported claim under the declared judging rule and zero otherwise. In the next display, $\mathbb{E}$ means an average, each $\sim$ means "sampled from," and $P_\Theta(\,\cdot \mid q)$ is the response distribution of model Θ under the fixed decoder; the dot is a placeholder for all possible responses. The population unsupported-response rate is

$$H(\Theta) = \mathbb{E}_{\substack{q\sim\mathcal{D}_{\rm eval} \\ Y\sim P_\Theta(\,\cdot\mid q)}}[Z(q,Y)]. \tag{A23}$$

In words: draw an evaluation case, generate a response, record either zero or one according to the judging rule, and average over repetitions. The probability law $P_\Theta(\,\cdot \mid q)$ includes the fixed decoder and its randomness. For deterministic greedy decoding it is concentrated on one response. A study that instead counts individual claims must redefine Z and must not treat several claims from the same response as independent cases.

For N held-out evaluation cases $q_1, \ldots, q_N$, let n index a case and let $Y_n^{(\Theta)}$ be the response generated by model Θ on that case. Define the measured rate $H_N(\Theta)$ to be the ordinary sample mean

$$H_N(\Theta) = \frac{1}{N}\sum_{n=1}^{N} Z(q_n, Y_n^{(\Theta)}). \tag{A24}$$

Every symbol in this quantity now has an observable role: N counts evaluation cases, each Z is zero or one, and $H_N \in [0,1]$ is their measured fraction.

For aligned parameters $\Theta_{\rm align}$ and matched control parameters $\Theta_{\rm ctrl}$, let $\Delta^{\rm align}_{\rm hall}$ denote the population difference and let its hatted version denote the finite-sample estimate. The superscript "align" is a descriptive label, not a power. Define

$$\begin{aligned} \Delta^{\rm align}_{\rm hall} &= H(\Theta_{\rm align}) - H(\Theta_{\rm ctrl}), \\ \widehat{\Delta}^{\rm align}_{\rm hall} &= H_N(\Theta_{\rm align}) - H_N(\Theta_{\rm ctrl}). \end{aligned} \tag{A25}$$

The control $\Theta_{\rm ctrl}$ must start from the same $\Theta_{\rm base}$ and be matched as closely as possible in data exposure, optimization and compute budget, evaluation prompts, grounding, decoding, and random seeds, except for the declared alignment intervention. Only when that intervention is the sole systematic difference can the contrast be attributed causally to it. One may set $\Theta_{\rm ctrl} = \Theta_{\rm base}$ only when "no post-training" is the explicit control condition. If the control receives a different form of post-training, that procedure and its data must be stated.

The hypothesis that alignment increases hallucination in the declared regime is $\Delta^{\rm align}_{\rm hall} > 0$; reduction corresponds to a negative value, and zero means no population difference on this measure. Nothing in the formalism fixes the sign. Coverage and abstention must be reported beside H_N, since universal refusal could lower the unsupported-response rate without adding knowledge. If the joint pairs of evaluation cases and generated responses, including decoder randomness, are independent draws from the fixed protocol, the law of large numbers explains why H_N approaches H as N grows. A finite study still requires uncertainty intervals and variation across training seeds. The equations therefore turn the proposed concern about alignment into a testable comparison; they do not assume its conclusion.

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